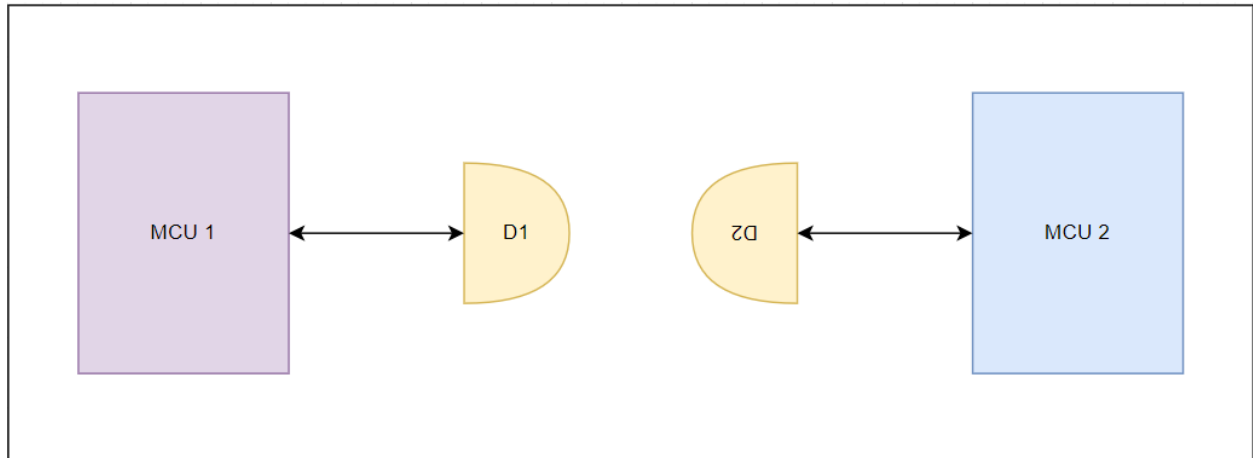




1. Basic Performance Tests

- **Range Testing:** Measure the maximum communication range (up to 100 mm) under different alignment conditions.
- **Data Rate Testing:** Evaluate the transceiver's performance at various data rates (125 Mb/s to 5 Gb/s).
- **Alignment Tolerance:** Test the impact of angular misalignment on communication reliability.

2. Custom Protocol Experiment

- Implement a custom protocol over the transceiver and measure latency, jitter, and data integrity.
- Use microcontrollers or FPGAs to interface with the transceiver via SMA connectors.

3. Environmental Testing

- Test the transceiver's performance across its operational temperature range (0°C to +85°C).
- Introduce optical interference (e.g., sunlight or other IR sources) to study noise immunity.

4. Mini Project: Wireless UART Communication

- Set up a simple wireless UART system using two boards.
- Send text messages between two microcontrollers over the optical wireless link.